Algebraic Techniques

Week 2D

Overview

Indices

Key Ideas - Zero Index Rule

- Negative Index Rules

- Fractional Index Rule

- Combined Index Rules

Key Idea – Zero Index Rule

Discussion Questions:

Why is $2^0 = 1$?

Consider : $2^2 \div 2^2 = 2^{2-2}$

$= 2^0$

Also

Consider : $\dfrac{2^2}{2^2} = \dfrac{\cancel{2} \times \cancel{2}}{\cancel{2} \times \cancel{2}}$

$= 1$

So $2^0 = 1$

Rule 6: Zero Index

To understand this rule, we can expand the following:

$$a^n \div a^n = a^{n-n} = a^0 \qquad \frac{\cancel{a^n}}{\cancel{a^n}} = 1$$

Combining both results we get:

Rule 6: $a^0 = 1$, where $a \neq 0$

***Examples*:**

$$4 \times 6^0 = 4 \times 1$$
$$= 4$$

$$\left(2^2 + \frac{1}{16}\right)^0 = 1$$

$$4\left(3ab^2\right)^0 + 1 = 4 \times 1 + 1$$
$$= 5$$

Note: *That 0^0 is not defined and is often referred to as "indeterminate".*

Exercises: Simplify and evaluate, where possible:

(a) $a \times x^0 + 3$ $= a \times 1 + 3$

$= a + 3$

(b) $\left(a \times x^0 + 3\right)^0 + 4$ $= 1 + 4$

$= 5$

(c) $\left(2^2 + \frac{1}{16}\right)^0$ $= 1$

Key Idea – Negative Index Rules

Rule 7: Negative powers

To understand this rule, we can expand the following:

$$a^2 \div a^5 = a^{2-5}$$
$$= a^{-3}$$

$$a^2 \div a^5 = \frac{\cancel{a} \times \cancel{a}}{\cancel{a} \times \cancel{a} \times a \times a \times a}$$
$$= \frac{1}{a^3}$$

Combining both results we get: $a^{-3} = \frac{1}{a^3}$

More generally, for any positive integer n,

$$a^{-n} = \overbrace{\frac{1}{a} \times \frac{1}{a} \times \cdots \times \frac{1}{a}}^{n \text{ terms}} = \frac{1}{\underbrace{a \times a \times \cdots \times a}_{n \text{ terms}}} = \frac{1}{a^n}$$

So, we end up with

$$\textbf{Rule 7: } \boldsymbol{a^{-n} = \frac{1}{a^n}} \textbf{ where } \boldsymbol{a \neq 0}$$

In particular, $$\boldsymbol{a^{-1} = \frac{1}{a}} \textbf{ where } \boldsymbol{a \neq 0}$$

***Examples:* Simplify the following, writing each answer with positive indices:**

$$x^{-4} = \frac{1}{x^4}$$

$$2x^{-3} = 2 \times \frac{1}{x^3} = \frac{2}{x^3}$$

$$3^{-2}a^3y^{-3} = 3^{-2} \times a^3 \times y^{-3} = \frac{1}{3^2} \times a^3 \times \frac{1}{y^3} = \frac{a^3}{3^2y^3}$$

$$x^2y^{-1} = x^2 \times y^{-1} = x^2 \times \frac{1}{y} = \frac{x^2}{y}$$

Exercises:

Use rules of indices to simplify the following expressions. Express each answer with positive indices and evaluate where possible:

(a) $(4x)^{-2} = \frac{1}{(4x)^2} = \frac{1}{4^2x^2} = \frac{1}{16x^2}$

(b) $4x^{-2} = \frac{4}{x^2}$

(c) $(-2a^3)^{-2} = \frac{1}{(-2a^3)^2} = \frac{1}{(-2)^2(a^3)^2} = \frac{1}{4a^{3\times2}} = \frac{1}{4a^6}$

(d) $7^{-2}a^3x^{-3} = \frac{1}{7^2} \times a^3 \times \frac{1}{x^3} = \frac{a^3}{7^2x^3}$

Special Case: Negative powers of numbers or variables divided

Discussion Questions:

Is $\frac{1}{2^{-5}} = 2^5$? Explain your answer. Yes

$$\frac{1}{2^{-5}} = \frac{1}{\frac{1}{2^5}} = 1 \div \frac{1}{2^5} = 1 \times \frac{2^5}{1} = 2^5$$

Is $\left(\frac{3}{2}\right)^{-1} = \frac{2}{3}$? Explain your answer. Yes

$$\left(\frac{3}{2}\right)^{-1} = \frac{1}{\left(\frac{3}{2}\right)} = 1 \div \frac{3}{2} = 1 \times \frac{2}{3} = \frac{2}{3}$$

To derive the special cases we can use rules 4 and 7:

$$\frac{1}{a^{-n}} = \frac{1}{\frac{1}{a^n}} \qquad \text{(rule 7)}$$

$$= 1 \div \frac{1}{a^n}$$

$$= 1 \times \frac{a^n}{1}$$

$$= a^n$$

$$\left(\frac{a}{b}\right)^{-n} = \frac{1}{\left(\frac{a}{b}\right)^n} \qquad \text{(rule 7)}$$

$$= \frac{1}{\frac{a^n}{b^n}} \qquad \text{(rule 4)}$$

$$= 1 \div \frac{a^n}{b^n}$$

$$= 1 \times \frac{b^n}{a^n}$$

$$= \frac{b^n}{a^n}$$

$$= \left(\frac{b}{a}\right)^n \qquad \text{(rule 4)}$$

$$\boldsymbol{\frac{1}{a^{-n}} = a^n \textbf{ where } a \neq 0}$$

$$\boldsymbol{\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n \textbf{ where } a \neq 0}$$

Examples: Use the rules for negative indices to simplify:

$$\frac{1}{2^{-3}} = 2^3 = 8$$

$$\left(\frac{3a}{2}\right)^{-3} = \left(\frac{2}{3a}\right)^3 = \frac{2^3}{(3a)^3} = \frac{8}{27a^3}$$

Exercises: Use the rules of indices to simplify the following expressions. Express each answer with positive indices and evaluate where possible.

(a) $\frac{1}{(4x)^{-2}} = \frac{(4x)^2}{1}$

$= (4x)^2$

$= 4^2x^2$

$= 16x^2$

(b) $\frac{4xy^{-1}}{z^{-2}} = \frac{4xz^2}{y}$

Key Idea – Fractional Index Rule

We define **fractional indices** as

$$a^{1/n} = \sqrt[n]{a}$$

We can then combine this with Rule 5 to see that for any integers *m* and *n*,

$$a^{m/n} = a^{(1/n)\times m} = (a^{1/n})^m = \left(\sqrt[n]{a}\right)^m$$

We could also go about this another way:

$$a^{m/n} = a^{m\times(1/n)} = (a^m)^{(1/n)} = \sqrt[n]{a^m}$$

So, we obtain another index rule:

Rule 8: $\boldsymbol{a^{m/n} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m}$

Previously, we worked with **surds**, which involve square roots and higher roots.
Fractional indices provide a way to write surds using **index notation**.
For example:

$$\sqrt{x} = x^{\frac{1}{2}} \text{ and } \sqrt[3]{x} = x^{\frac{1}{3}}$$

In other words, **surds are a special case of fractional indices**, and all the **index laws** we have already learned continue to apply.

***Note*:**

- *When using fractional indices, care must be taken to avoid meaningless expressions; in other words, care must be taken to see that the appropriate root exists. For example,* $2^{1/2}$ *exists as it is a real number, but* $(-2)^{1/2}$ *is not.*
- *For most fractional indices it is more convenient to use* $\left(\sqrt[n]{a}\right)^m$ *to evaluate* $a^{m/n}$.

$a^{1/n}$ *is also written as* $\sqrt[n]{a}$ *and we read both expression as the* ***nth root of a****.*

Examples: Write the following expressions in index form:

$$\sqrt[5]{a} = a^{\frac{1}{5}} \qquad \sqrt{3x-4} = (3x-4)^{\frac{1}{2}}$$

$$\frac{1}{2\sqrt[3]{(3a+1)^4}} = \frac{1}{2(3a+1)^{\frac{4}{3}}}$$

$$= \frac{(3a+1)^{\frac{-4}{3}}}{2}$$

Exercises: Write the following expressions in index form:

(a) $\dfrac{3}{4\sqrt{(b-2)^3}}$ $= \dfrac{3}{4(b-2)^{\frac{3}{2}}}$ or $\dfrac{3(b-2)^{-\frac{3}{2}}}{4}$

(b) $\dfrac{x^2}{\sqrt[3]{x}}$ $= \dfrac{x^2}{x^{\frac{1}{3}}} = x^{2-\frac{1}{3}} = x^{\frac{5}{3}}$

***Examples*:**

Write each of the following expressions without fractional or negative indices:

$$a^{p/4} = \sqrt[4]{a^p} = \left(\sqrt[4]{a}\right)^p$$

$$\left(xy^3\right)^{5/3} = x^{5/3}\left(y^3\right)^{5/3} = \sqrt[3]{x^5}\,y^{5\times(5/3)} = \sqrt[3]{x^5}\,y^5$$

$$\left(xy^3\right)^{-5/3} = \frac{1}{\left(xy^3\right)^{5/3}} = \frac{1}{x^{5/3}\left(y^3\right)^{5/3}} = \frac{1}{x^{5/3}y^{3\times5/3}} = \frac{1}{x^{5/3}y^5} = \frac{1}{\sqrt[3]{x^5}\,y^5}$$

$$(a+b)^{\frac{2}{7}} = \sqrt[7]{(a+b)^2}$$

$$\frac{2(3a+1)^{\frac{-2}{7}}}{5} = \frac{2}{5(3a+1)^{\frac{2}{7}}} = \frac{2}{5\sqrt[7]{(3a+1)^2}}$$

Exercises: Write the following expressions without fractional or negative indices:

(a) $\left(xy^3\right)^{\frac{4}{5}}$ $= \sqrt[5]{(xy^3)^4}$

$= \sqrt[5]{x^4(y^3)^4}$

$= \sqrt[5]{x^4y^{3\times4}}$

$= \sqrt[5]{x^4y^{12}}$

(b) $\frac{1}{2}(a+b)^{\frac{-5}{3}}$

$= \frac{1}{2} \times \frac{1}{(a+b)^{\frac{5}{3}}}$

$= \frac{1}{2} \times \frac{1}{\sqrt[3]{(a+b)^5}}$

$= \frac{1}{2\sqrt[3]{(a+b)^5}}$

Key Idea – Combined Index Rules

We will now combine all the rules and the special cases we have seen to simplify expressions.

As a summary, if a and b are real numbers and m and n are integers, then

- $a^n \times a^m = a^{n+m}$
- $a^n \div a^m = a^{n-m}$
- $(ab)^n = a^n b^n$
- $\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$ (provided $b \neq 0$)
- $\left(a^n\right)^m = a^{n\times m}$
- $a^0 = 1$ (provided $a \neq 0$)
- $a^{-n} = \dfrac{1}{a^n}$ (provided $a \neq 0$)
- $\dfrac{1}{a^{-n}} = a^n$ (provided $a \neq 0$)
- $\left(\dfrac{a}{b}\right)^{-n} = \left(\dfrac{b}{a}\right)^n$ (provided $a \neq 0$)
- $a^{m/n} = \sqrt[n]{a^m} = \left(\sqrt[n]{\mathrm{a}}\right)^{\mathrm{m}}$

Examples: Write each expression in simplest index form, using positive indices where possible.

$$\frac{(x^3)^{-2}\times x^{-3}}{x^2}=\frac{x^{3\times -2}\times x^{-3}}{x^2}=\frac{x^{-6}\times x^{-3}}{x^2}=x^{-6-3-2}=x^{-11}=\frac{1}{x^{11}}$$

$$\left(a^2b^3\right)^{-1}\times ab^{-2}\times\left(c^{-1}\right)^{-1}=a^{-2}b^{-3}\times ab^{-2}\times c^1=a^{-2+1}b^{-3-2}\times c^1=a^{-1}b^{-5}\times c^1=\frac{c}{ab^5}$$

$$\frac{a^2b^3\times a^2b^2\times b}{b^4\times ab}=\frac{a^{2+2}b^{3+2+1}}{b^{4+1}a^1}=\frac{a^4b^6}{b^5a}=a^{4-1}b^{6-5}=a^3b$$

$$\frac{\left(xy^2\right)^3\times\left(x^2y\right)^2}{\left(x^2y^2\right)^2}=\frac{x^3\left(y^2\right)^3\times\left(x^2\right)^2y^2}{\left(x^2\right)^2\left(y^2\right)^2}=\frac{x^3y^{2\times 3}\times x^{2\times 2}y^2}{x^{2\times 2}y^{2\times 2}}=\frac{x^3y^6\times x^4y^2}{x^4y^4}=\frac{x^3y^6\times y^2}{y^4}=x^3y^{6+2-4}=x^3y^4$$

$$\frac{3^{n-2}\times 9^{n+1}}{81^{n-1}}=\frac{3^{n-2}\times(3^2)^{n+1}}{(3^4)^{n-1}}=\frac{3^{n-2}\times 3^{2n+2}}{3^{4n-4}}=\frac{3^{(n-2)+(2n+2)}}{3^{4n-4}}=\frac{3^{3n}}{3^{4n-4}}=3^{3n-(4n-4)}=3^{4-n}=\frac{1}{3^{n-4}}\quad\text{or}\quad\frac{81}{3^n}$$

$$(a^2-2b^{-1})(a^{-2}-b)=\left(a^2-\frac{2}{b}\right)\left(\frac{1}{a^2}-b\right)=a^2\times\frac{1}{a^2}-a^2\times b-\frac{2}{b}\times\frac{1}{a^2}+\frac{2}{b}\times b=1-a^2b-\frac{2}{a^2b}+2=3-a^2b-\frac{2}{a^2b}$$

$$\frac{x^{-1}+y^{-1}}{x+y}=\frac{\frac{1}{x}+\frac{1}{y}}{x+y}=\frac{\frac{y+x}{xy}}{x+y}=\frac{y+x}{xy}\div(x+y)=\frac{y+x}{xy}\times\frac{1}{x+y}=\frac{\cancel{x+y}}{xy}\times\frac{1}{\cancel{x+y}}=\frac{1}{xy}$$

$$x^{1/2}\times x^{1/3}=x^{1/2+1/3}=x^{5/6}$$

$$a^3\div a^{-\frac{1}{2}}=a^{3-(-\frac{1}{2})}=a^{3+\frac{1}{2}}=a^{\frac{7}{2}}$$

$$\left(2a^{\frac{1}{3}}\right)^{-3}=2^{-3}\left(a^{\frac{1}{3}}\right)^{-3}=\frac{1}{2^3}a^{\frac{1}{3}\times(-3)}=\frac{1}{8}a^{-1}=\frac{1}{8a}$$

$$\left(8m^{-2}n^4\right)^{1/2}=8^{1/2}\left(m^{-2}\right)^{1/2}\left(n^4\right)^{1/2}=\sqrt{8}m^{-2\times1/2}n^{4\times1/2}=\sqrt{8}m^{-1}n^2=\frac{\sqrt{8}n^2}{m}=\frac{2\sqrt{2}n^2}{m}$$

$$\frac{\sqrt{a^3x^3}}{\sqrt[3]{a^2x^3}}=\frac{\left(a^3x^3\right)^{\frac{1}{2}}}{\left(a^2x^3\right)^{\frac{1}{3}}}=\frac{a^{3\times\frac{1}{2}}x^{3\times\frac{1}{2}}}{a^{2\times\frac{1}{3}}x^{3\times\frac{1}{3}}}=\frac{a^{\frac{3}{2}}x^{\frac{3}{2}}}{a^{\frac{2}{3}}x}=a^{\frac{3}{2}-\frac{2}{3}}x^{\frac{3}{2}-1}=a^{\frac{5}{6}}x^{\frac{1}{2}}$$

$$\left(x^{\frac{1}{2}}-x^{-\frac{1}{2}}\right)^2=\left(x^{\frac{1}{2}}\right)^2-2\times x^{\frac{1}{2}}\times x^{-\frac{1}{2}}+\left(x^{-\frac{1}{2}}\right)^2=x^{\frac{1}{2}\times2}-2\times x^{\frac{1}{2}+\left(-\frac{1}{2}\right)}+x^{-\frac{1}{2}\times2}=x^1-2x^0+x^{-1}=x-2+\frac{1}{x}$$

Exercises: Write each expression in simplest index form, using positive indices where possible.

(a) $(xy)^3 \times \left(x^3\right)^2 \div x^2y^2 = x^3y^3 \times x^{3\times2} \div x^2y^2$

$= x^3y \times x^6 \div x^2y$

$= x^9y \div x^2y$

$= x^{9-2}y^{1-1}$

$= x^7y^0 \quad = x^7$

(b) $\dfrac{2xy^{-2} \times 8\left(xy^{-3}\right)^2}{4x^{-2}y} = \dfrac{2xy^{-2} \times 8x^2y^{-3\times2}}{4x^{-2}y}$

$= \dfrac{2\times2\times\cancel{4}\times x^{1+2}\cdot y^{-2+(-6)}}{\cancel{4}\times x^{-2}\times y}$

$= \dfrac{4x^3y^{-8}}{x^{-2}y}$

$= \dfrac{4x^3x^2}{y^8y}$

$= \dfrac{4x^{3+2}}{y^{8+1}}$

$= \dfrac{4x^5}{y^9}$

(c) $\dfrac{10^n - 4^n}{5^n - 2^n} = \dfrac{(2\times5)^n - (2\times2)^n}{5^n - 2^n}$

$= \dfrac{2^n \times 5^n - 2^n \times 2^n}{5^n - 2^n}$

$= \dfrac{2^n\cancel{(5^n - 2^n)}}{\cancel{(5^n - 2^n)}} \quad = 2^n$

(d) $6^a \times 3^b \times 2^2 = (3\times2)^a \times 3^b \times 2^2$

$= 3^a \times 2^a \times 3^b \times 2^2$

$= 3^{a+b}\ 2^{a+2}$

(e) $x^3 \times \left(x^{\frac{2}{3}}\right)^2 \div x^{-\frac{4}{5}}$

$$= x^3 \times x^{\frac{2}{3}\times 2} \div x^{-\frac{4}{5}}$$

$$= x^{3+\frac{4}{3}} \div x^{-\frac{4}{5}}$$

$$= x^{\frac{13}{3}} \div x^{-\frac{4}{5}}$$

$$= x^{\frac{13}{3}-\left(-\frac{4}{5}\right)}$$

$$= x^{\frac{13}{3}+\frac{4}{5}} = x^{\frac{77}{15}}$$

(f) $\dfrac{\left(\sqrt[5]{a}\right)^3 \times a^{\frac{-1}{2}}}{a^{\frac{4}{5}} \div \sqrt{a^{-3}}}$

$$= \frac{a^{\frac{3}{5}} \times a^{-\frac{1}{2}}}{a^{\frac{4}{5}} \div a^{-\frac{3}{2}}}$$

$$= \frac{a^{\frac{3}{5}+\left(-\frac{1}{2}\right)}}{a^{\frac{4}{5}-\left(-\frac{3}{2}\right)}}$$

$$= \frac{a^{\frac{1}{10}}}{a^{\frac{22}{10}}} = a^{\frac{1}{10}-\frac{22}{10}} = a^{-\frac{21}{10}} = \frac{1}{a^{\frac{21}{10}}}$$